Yarrow System Architecture & Processing Report

1. Overview & Ingestion Pipeline

Yarrow is a next-generation document processing platform engineered to parse, index, and analyze complex unstructured PDF files. When a user uploads a document via the web dashboard or imports directly through connected cloud storage integrations (such as Google Drive), the ingestion subsystem allocates an asynchronous worker task to orchestrate document layout analysis and optical character recognition (OCR).

Figure 1: High-level architectural flowchart of Yarrow document processing worker pipeline.

As demonstrated in Figure 1, source files move through a decoupled processing pipeline where worker nodes execute isolated containerized analysis tasks. Extracted content is decomposed into granular document regions - specifically disambiguating between continuous prose paragraphs, structural headers, embedded figures, and tabular matrices.

Each extracted bounding box is calculated relative to normalized page dimensions, ensuring that spatial coordinate mappings remain pinpoint accurate regardless of client display resolutions or zoom scaling factors.

2. Benchmark Performance & Layout Recognition

Empirical evaluations of Yarrow's parsing engine across diverse academic and financial datasets demonstrate robust throughput and low error rates. Table 1 summarizes throughput metrics collected across standard test document corpora, highlighting extraction accuracy across heterogeneous document layouts.

Corpus ID	Document Type	Pages	Processing Time	OCR Accuracy	Layout Status
DOC-101	Academic Article	14	2.85s	99.4%	Verified
DOC-102	Financial Balance Sheet	6	1.42s	98.9%	Verified
DOC-103	Technical Specification	22	4.15s	99.1%	Verified
DOC-104	Multi-Column Newsletter	4	0.98s	97.8%	Verified
DOC-105	Scanned Legal Contract	8	2.10s	98.5%	Verified

Table 1: Benchmark extraction performance and OCR validation metrics across test document formats.

3. Concluding Remarks & Inspection Workflows

By indexing bounding box metadata directly with source page numbers, Yarrow empowers collaborative teams to cross-examine extracted textual elements against the underlying document canvas in real time. Clicking on any extracted paragraph or table row immediately refocuses the viewport to the corresponding source region, eliminating manual visual scanning and enabling frictionless validation.

Furthermore, access control rules guarantee that processed assets and their associated bounding coordinate schemas remain strictly isolated to authorized project collaborators.